

From Leisure to Loyalty

Behavioral
Segmentation of
Divvy Bike-Share
Riders

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Business Problem

How do riding behaviors differ between casual riders and members across ride duration, timing, frequency, and station usage?

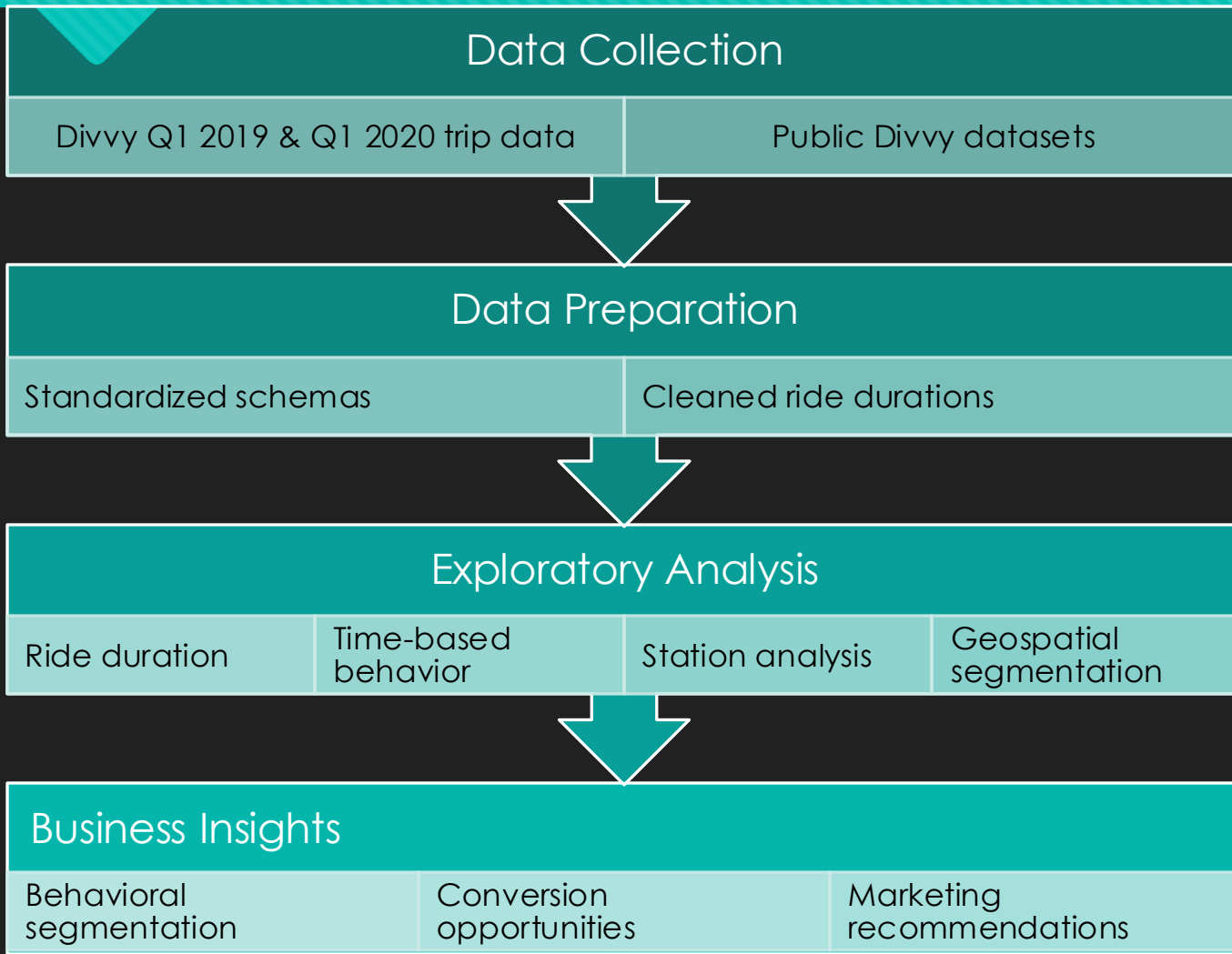
Data Sources

- Divvy Trips 2019 Q1
- Divvy Trips 2020 Q1
- public historical trip data
- combined after schema normalization

Data Cleaning & Preparation

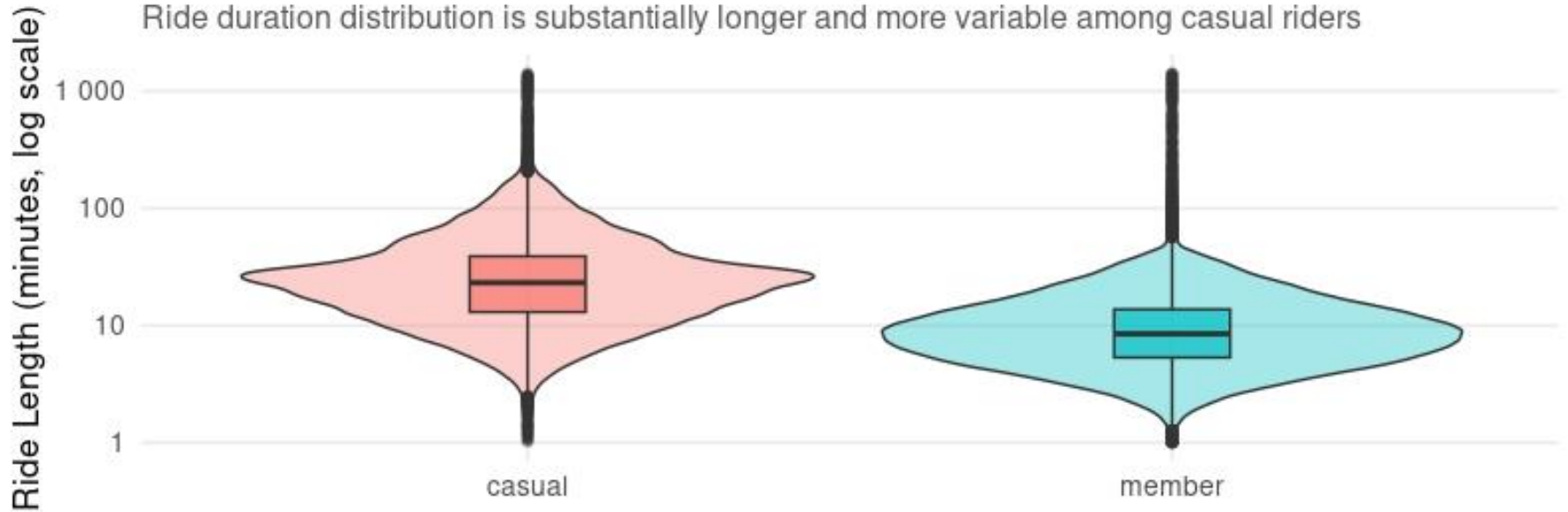
- standardized columns
- recoded Customer / Subscriber
- harmonized datatypes
- datetime parsing
- ride_length creation
- weekday/hour creation
- removed rides <1 min
- removed rides >24 hrs.
- stacked datasets with `bind_rows()`

Methodology



Casual Riders Take Significantly Longer Trips

Ride duration distribution is substantially longer and more variable among casual riders



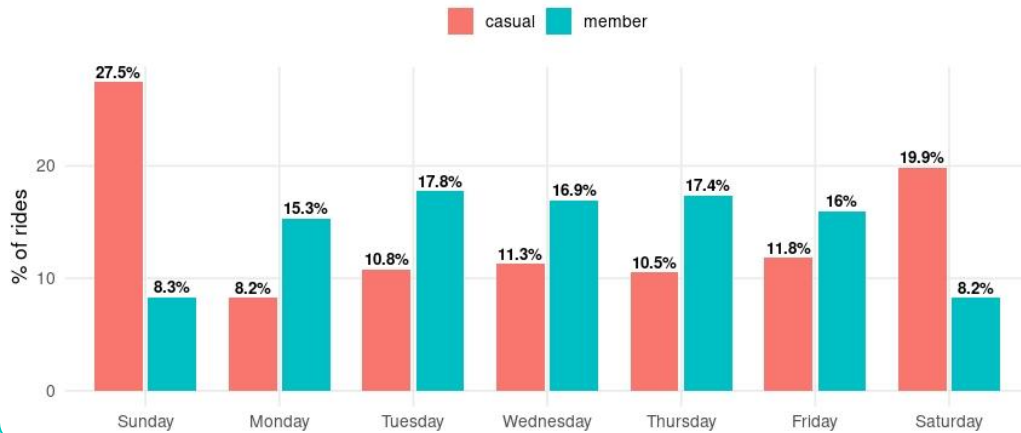
Key Finding 1: Ride Duration

- Casual riders tend to take longer and less consistent rides, while members take shorter, more routine trips.

Key Finding 2: Weekly Patterns

Weekend vs Weekday Usage Shows Clear Behavioral Segmentation

Casual riders cluster on weekends, while members ride primarily during the workweek

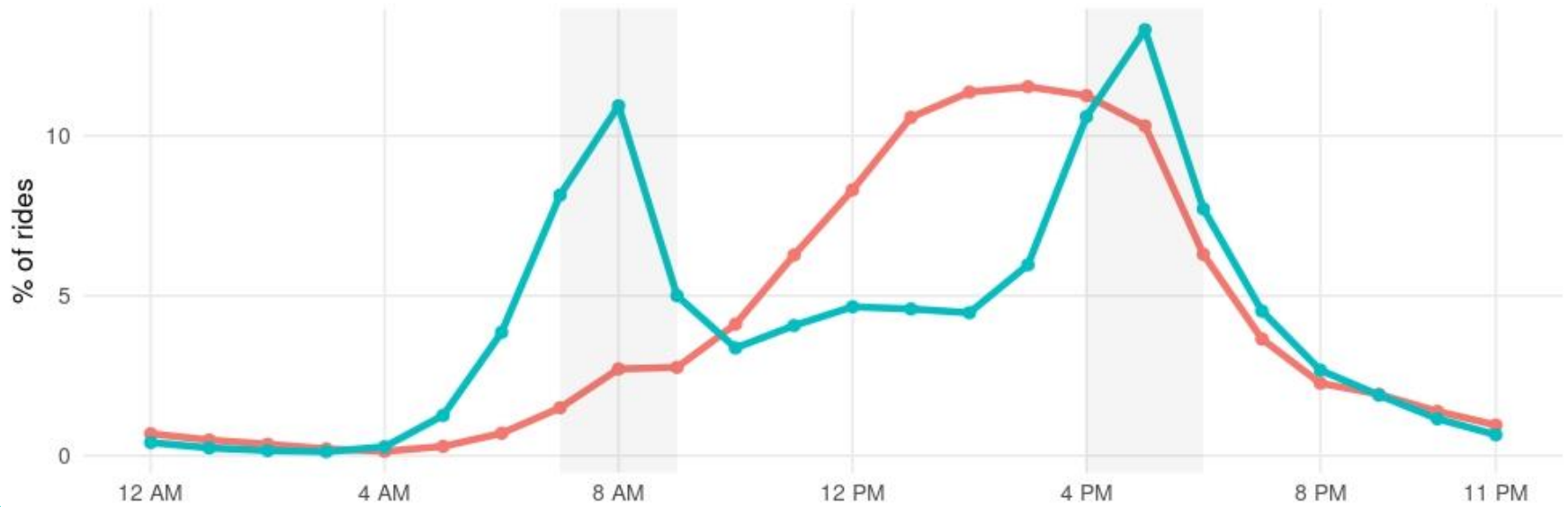


- While the strong weekend usage pattern of casual riders shows that nearly half of the rides occur on Saturdays and Sundays, suggesting primarily recreational usage; the members ride most frequently during weekdays, especially Tuesday through Thursday, indicating routine commuting behavior.

Members Follow a Commuting Rhythm; Casual Riders Follow a Leisure Rhythm

Commute peaks appear in morning and evening for members, while casual ridership peaks mid-afternoon

casual member

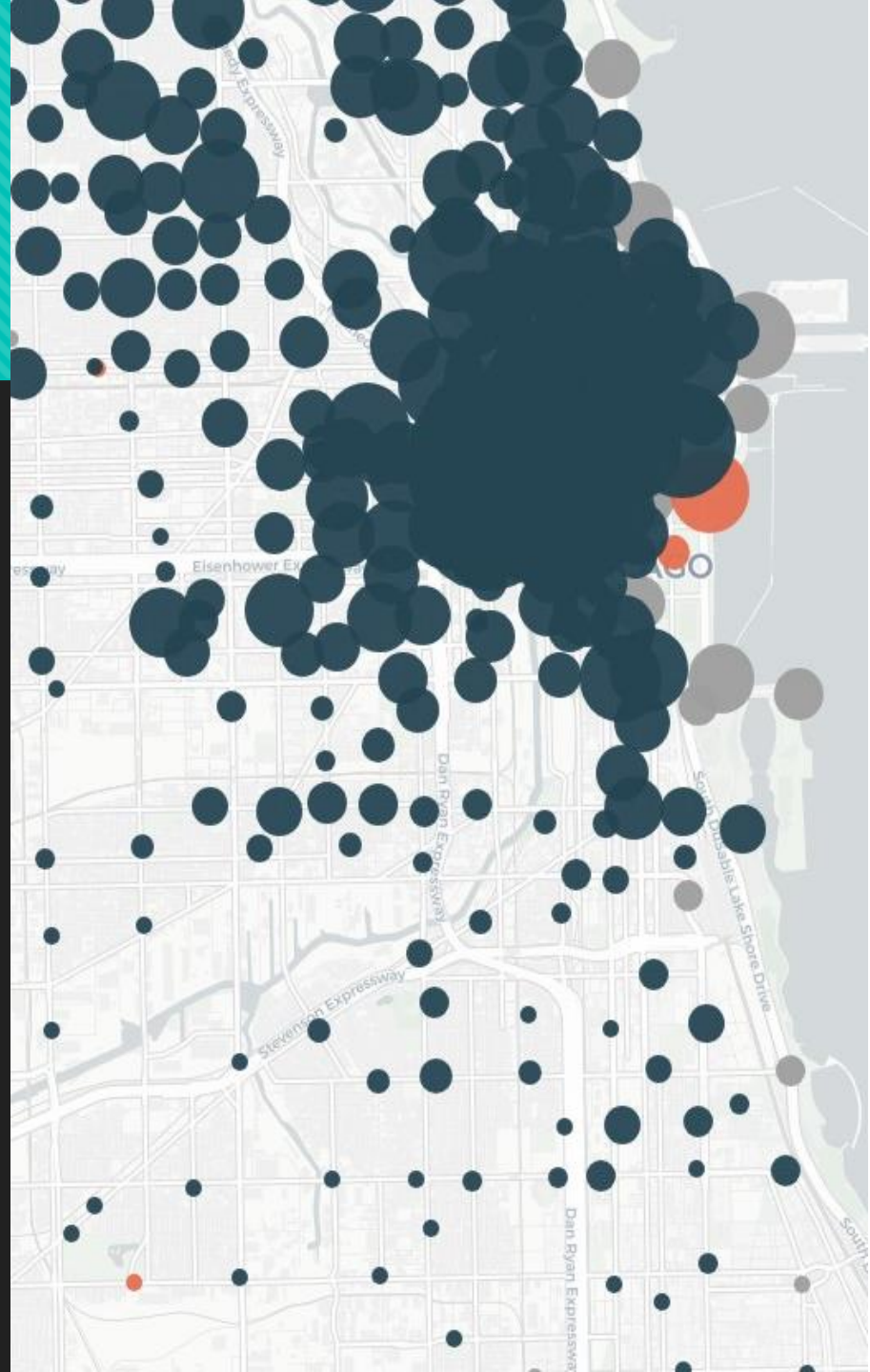


Key Finding 3: Hourly Patterns

- Members with strong ride spikes during morning commuting hours and again during evening commuting hours suggest that members primarily use Divvy as a transportation solution for work related commuting. Casual riders, by contrast, show gradual increase in trips in

Key Finding 4: Top Stations

- Casual riders cluster around Chicago's lakefront, parks, attraction-adjacent stations, and recreation-focused corridors, while members concentrate around downtown business districts, office centers, and transit hubs. Geospatial mapping highlights a clear separation between leisure-oriented and commuter-oriented usage zones.



*Map data from 2020 dataset only

Recommendations



**1. PILOT A WEEKEND-
FOCUSED MEMBERSHIP
OR RIDE BUNDLE**



**2. TARGET HIGH-
CONCENTRATION LEISURE
STATIONS FOR
CONVERSION
CAMPAIGNS**



**3. DEVELOP
PERSONALIZED
ENGAGEMENT
CAMPAIGNS FOR HIGH-
FREQUENCY CASUAL
RIDERS**

Limitations

- only Q1 data
- no demographic data
- cannot directly identify repeat individual casual riders
- station context inferred

Next Steps

- analyze full-year seasonality
- geospatial mapping
- customer clustering
- A/B testing campaigns